IV. (points) In thermodynamic cycles, engines take heat from a hot source, output part as work and reject the rest to a cold source. Heat pumps use external work to move cold-source heat to a hot source, together with that work converted to heat.

By the second law, if either working substance contacts only two reservoirs at ,

absorbed heats obey . Positive and negative values mean

heat absorption from, and heat rejection to, a reservoir, respectively.

A heater uses a boiler at to heat a room directly, maintaining a constant room

temperature , above outdoors at . Using the same energy source and the machines,

what is the theoretical maximum fractional energy saving compared with direct heating?